RANIA A. HAMID

New Carrollton, MD | ran.hamid@gmail.com | +1 (240) 381-9984 | US Citizen | Eligible for US Security Clearance

Data Scientist | Applied Machine Learning & Analytics for Government Systems

Data scientist with 4.5 years of experience in applied analytics, machine learning, and data engineering, supported by 7+ years in operational and program support within complex organizational environments. Proven record of delivering enterprise impact, including preventing $790K in government waste, resolving $200M in audit findings, and improving reporting efficiency by 98% through automated data pipelines. Skilled in Python, R, and SQL with expertise in predictive modeling, statistical analysis, neural networks, and generative AI architectures. Experienced translating complex operational data into actionable insights, scalable analytics solutions, and decision-support systems for leadership.

CORE EXPERTISE

Machine Learning & AI: Predictive Modeling | Generative AI & Transformers | Neural Networks | AI Evaluation & Model Testing | **Data Analytics & Statistics:** Data Analysis & Visualization | Hypothesis Testing | Statistical Modeling | Fraud & Waste Detection | **Data Engineering & Pipelines:** Data Pipeline Development | ETL/ELT | Process Automation | Data Quality Management | Data Governance | **Programming & Tools:** Python (Pandas, NumPy, Scikit-learn) | R (Tidyverse, Caret) | SQL | TensorFlow | PyTorch | PySpark | Databricks | **Analytics & BI Platforms:** Tableau | IBM Cognos Analytics | Smartsheet | **Business & Government Operations:** Audit Support | Financial Accountability | Government Operations | Regulatory Compliance | **Collaboration & Leadership:** Stakeholder Collaboration | Cross-Functional Team Leadership | Enterprise Design Thinking | **Development & Data Standards:** Git/GitHub | XML/XBRL

EDUCATION & PROFESSIONAL DEVELOPMENT

M.S. Data Analytics | University of Maryland Global Campus **Expected 08/2026**

Relevant Coursework: Data Management, Data Visualization, Ethical AI, Predictive Analytics, Machine Learning, Deep Learning & Neural Networks, Reinforcement Learning, Generative Adversarial Networks, Transformers

Certificate, Unleashing the Power of AI Agents | IBM **03/2026**

Certificate, Introduction to Retrieval Augmented Generation | IBM **03/2026**

Certificate, Enterprise Design Thinking Practitioner | IBM **01/2026**

Certificate Data Science | Woz-U **05/2022**

Relevant Coursework: Statistical Process Control, Agile/SCRUM, Big Data Processing, Statistics, Survey Design and Analysis

B.A. Linguistics, Minor in Japanese | University of Maryland - College Park **05/2012**

Relevant Coursework: Calculus I & II, Syntax I & II, Pragmatics

PROFESSIONAL EXPERIENCE

Business Analyst → Senior Data Analyst | The Midtown Group (Contract), Washington, DC **06/2022 – 04/2024**

Contract role supporting Washington DC Office of Contracting and Procurement | 40 hours per week

- Architected and deployed enterprise data pipelines integrating GovDeals (7 active + 2 legacy accounts, 100K+ records), BarCloud (100K+ records across 5 warehouses), Quickbase, Smartsheet, AssetTiger, and Microsoft SQL Server, unifying 10+ years of procurement, financial, and inventory data (2011–2024) supporting ~$472M in government assets.
- Engineered automated ETL pipelines (Python: pandas, NumPy; SQL) with weekly/monthly refresh cadences across 10+ datasets, eliminating manual data aggregation and reducing reporting latency from hours to <1 minute (~98% improvement).
- Developed and cross-validated multivariate regression models (R: caret, lmtest, car, fastR2) using backward stepwise feature selection, improving revenue forecast accuracy by ~59% vs. baseline (error reduced from $6.5M to ~$1M).
- Built predictive models for contract burn rate forecasting (R/Python) in collaboration with finance stakeholders, enabling proactive budget reallocation and preventing premature contract exhaustion by ~3 months.
- Designed and deployed executive dashboards in Tableau and Smartsheet, delivering near real-time insights to director-level leadership and analysts and automating high-frequency reporting workflows.
- Applied statistical analysis and operational optimization (FIFO inventory policy) with logistics and warehouse teams, preventing ~$790K in immediate waste and reducing risk of multimillion-dollar losses from expiring PPE and COVID-19 inventory.
- Standardized and integrated procurement, financial, and inventory datasets (P-Card, PASS POs, vouchers, warehouse transactions) using SQL Server, enabling transaction-level traceability and supporting resolution of $200M+ in audit findings.
- Authored and implemented data governance SOPs introducing procurement ID as a system-wide unique key, closing critical traceability gaps and enabling 3-way reconciliation across procurement, finance, and distribution systems.

- Led cross-functional data science initiatives with procurement, finance, logistics, and public health stakeholders, translating business requirements into scalable analytical solutions supporting mission-critical operations.
- Developed reusable data science workflows and modular codebases (Python, R: dplyr, tidyr, ggplot2, shiny) adopted by 3–5 analysts, standardizing modeling, visualization, and reporting automation practices.
- Delivered executive-level insights and policy recommendations by translating statistical outputs into actionable strategies, directly influencing procurement planning, contract oversight, and resource allocation decisions.

SPECIALIZED EXPERIENCE

AI Experiential Learning Lab & Hackathon | IBM SkillsBuild (Online) **01/2026 – Present**

Applied Data Science & Analytics Training | 20-30 hours per week

- Led a cross-functional team of five first-time hackathon participants for the IBM SkillsBuild AI Experiential Learning Lab, recruiting members across AI engineering, systems engineering, business management, and grant administration; managed all team operations including document library buildout, deliverable submission, working session facilitation, and cross-functional coordination
- Led team in the design and prototyping of a two-stage agentic AI system built on IBM Watsonx that standardizes dynamic load disclosure between data center operators and grid operators, addressing a gap NERC formally identified as high impact and high likelihood for bulk power system reliability, integrating NLP-based document parsing, structured intake, and a pattern-learning early warning system for transmission planners

Data Science Bootcamp | Woz U, LLC (Online) **03/2021 – 05/2022**

Applied Data Science & Analytics Training | 25 hours per week

- Applied ML/DL & Statistical Modeling (Python/R): Completed intensive training in supervised/unsupervised methods, algorithm selection, model evaluation and cross-validation to support analytical integrity
- Data Engineering & Big Data Tech (SQL/AWS): Executed SQL-based extraction, cleaning, and normalization to ensure rigorous quality assurance for large-scale analytical modeling; gained proficiency in Big Data technologies (Hadoop, Spark, Pig, Hive) and AWS cloud environments
- Statistical Analysis & Inferential Modeling: Executed complex inferential statistics, including ANOVAs, ANCOVAs, MANOVAs, and MANCOVAs, to validate experimental hypotheses and conduct survey analysis. Ensured model integrity by rigorously testing assumptions of homogeneity of variance, homoscedasticity, and independence; utilized G-Power for power analysis and sample size determination.

SAMPLE TECHNICAL PROJECTS

Neural Network for Sentiment Analysis | TensorFlow, Python, NLP

Built and evaluated feed-forward neural network using TensorFlow to classify sentiment in text data from GLUE SST-2 dataset; built human review trigger alert

Generative Adversarial Networks (GANs) | TensorFlow, DCGAN, WGAN-GP

Developed and tuned GAN Architectures (DCGAN, WGAN-GP) to synthesize complex 64x64 imagery, implementing gradient penalties to resolve vanishing gradients and mode collapse. Evaluated using FID and systematic hyperparameter ablation studies to improve image distribution matching. Framed research findings as a proxy for generating synthetic load profiles for transmission planners

Netflix SEC Data Pipeline | Python, ETL, Data Engineering

Engineered automated ETL pipeline to extract financial data from SEC EDGAR filings

Grid Monte Carlo Simulator| PySpark, Python, Big Data

(In progress) Built probabilistic risk assessment framework using Monte Carlo methods to calculate loss of load probability and capacity shortfall metrics using EIA generator, weather, and planned data center infrastructure datasets

EARLY CAREER (CONDENSED)

Sr Administrative Assistant, Legal | NERC, Washington, DC **05/2020 – 03/2022**
Sr Administrative Assistant, Enforcement | Randstad Staffing, Inc., Washington, DC **02/2020 – 05/2020**
Data Maintenance Coordinator | RP Staffing/CASE | Washington, DC **03/2019 – 12/2019**
Executive Assistant | RP Staffing/AHS | Alexandria, VA **12/2018 – 03/2019**
Member Benefits and Services Support| RP Staffing/AAMC | Washington, DC **09/2018 – 11/2019**
Membership Support Specialist | RP Staffing/AAPS | Arlington, VA **07/2018 – 09/2018**
Floor Manager | Diyanet Center of America – Turkish Bath & Sports Center | Lanham, MD **09/2017 – 02/2018**
Licensed Insurance Representative | Transamerica Agency Network | Falls Church, VA **03/2017 – 08/2017**
Licensed Insurance Agent | Banker's Life & Casualty | Baltimore, MD **09/2016 – 02/2017**

Development Assistant | T.H.I.S. (now Cultural Exchange) for Diplomats | Washington, DC 04/2014 – 07/2016